



JURONG JUNIOR COLLEGE

Preliminary Examination 2009

MATHEMATICS Higher 1

8863

18 August 2009

3 hours

Additional materials: Writing Material
List of formulae (MF15)
Cover page

READ THESE INSTRUCTIONS FIRST

Write your name, class, and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded the need of clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 100.

At the end of the paper, fasten all your work securely together, with the cover page in front.

This question paper consists of **7** printed pages and **1** blank page.

[Turn over

Section A: Pure Mathematics [40 marks]

1. Solve the equation $8\log_3 \sqrt{3} + \log_2(3x+4) = 2\log_2 10$. [3]

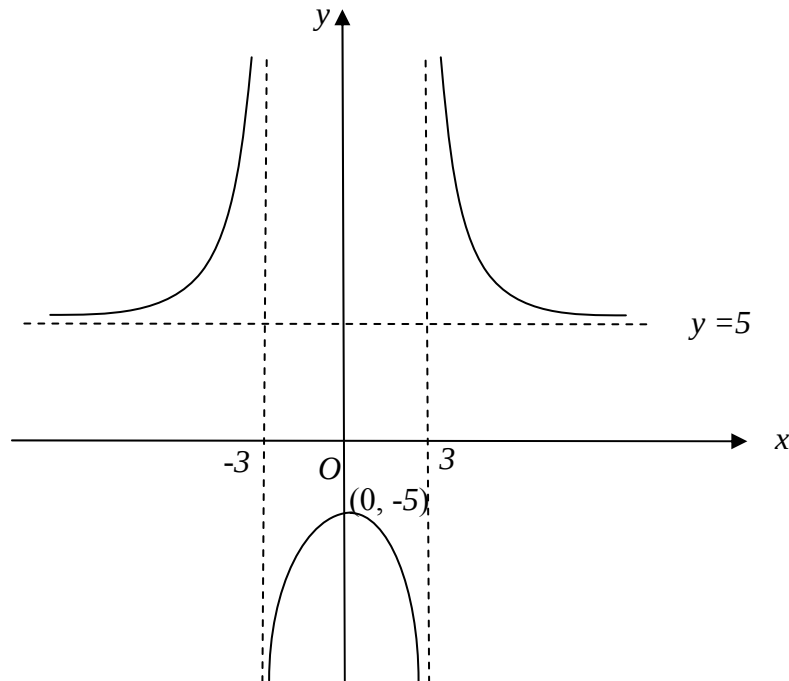
2. A cylindrical container of radius 2 cm contains water to a depth of 4 cm. The water is then poured at a steady rate into an inverted conical container with its axis vertical. After t seconds, the depth of water in this container is x cm and the volume V cm³ of water that has been transferred is given by $V = \frac{1}{3}\pi x^3$.

Given that all the water is transferred in 2 seconds, find

(i) $\frac{dV}{dt}$ in terms of π , [1]

(ii) the rate at which x is increasing at the moment when $x = 1$ cm. [2]

3. The graph of $y = \frac{ax^2 + b}{x^2 - c}$ is shown below, where a, b, c are constants. The curve passes through $(0, -5)$.



(i) Find the values of a, b and c . [4]

(ii) State the range of values of k for which the equation $\frac{ax^2 + b}{x^2 - c} = k$ has 2 real roots. [2]

4. The functions f and g are defined as follows:

$$f: x \rightarrow x^2 + 2, \quad x \geq 0,$$

$$g: x \rightarrow \ln x, \quad x \geq 1.$$

- (i) Show that $fg(x)$ exists. [1]
- (ii) Find, in similar form, the inverse function f^{-1} . [3]
- (iii) Sketch the graph of $f^{-1}f$. [2]
5. (a) On the same axes, sketch the graphs of $y = \sin x$ and $y = 2 \cos 3x$ for $0 \leq x \leq \pi$, labeling each graph carefully. [2]
- Hence solve the inequality $\cos 3x - \frac{1}{2} \sin x > 0$ for $0 \leq x \leq \pi$. [2]
- (b) Find the exact values of k for which $y = -x + k$ intersects the curve $y = 2 + \frac{3}{x-1}$ at exactly one point. [4]
6. (a) Find, simplifying your answers,
- (i) $\int (3x-4)^{-1} dx$, [2]
- (ii) $\int (6e^{\frac{x}{2}})(e^{\frac{x}{4}}) dx$. [2]
- (b) The equation of a curve is given by $y = 4x^2 + 1$. Find, in terms of p , the equation of the tangent to the curve at the point $A(p, 4p^2 + 1)$. [3]
- If the normal of the curve at point A meets the x -axis at C and F is the foot of the perpendicular from point A to the x -axis, find the length of CF in terms of p . [5]
- Region B is bounded by the curve $y = 4x^2 + 1$, the x -axis, the y -axis and the line AF . Find the area of B in terms of p . [2]

Section B: Statistics (60 marks)

7. Two ordinary fair dice, one red and one blue, are to be rolled once.
- (i) Find the probabilities of the following events:
 Event A: the number showing on the red dice will be a 5 or a 6,
 Event B: the total of the numbers showing on the two dice will be 4,
 Event C: the total of the numbers showing on the two dice is divisible by 2. [3]
 - (ii) State, with a reason, which of the two of the events A, B and C are mutually exclusive. [1]
 - (iii) Show that the events A and C are independent. [2]
8. Tom travels to school either by train, bus or taxi with the probability of 30%, 50% and 20% respectively. When Tom travels by train, there is a probability of 10% that he will be late for school. The corresponding percentages that Tom will be late for school when he travels by bus or taxi are 15% and 6% respectively. Calculate the probability that Tom
- (i) travels to school by bus and will not be late for school, [1]
 - (ii) will be late for school, [2]
 - (iii) will not be late for school for three consecutive days. [2]
- Given that Tom will be late for school, find the probability that he does not travel to school by train. [2]
9. In a large batch of items from a production line, it was found that the probability that an item is faulty is p , where $0 < p < 0.5$.
- A random sample of 15 items is taken and the random variable X is the number of faulty items. Given that $P(X = 8) = 0.08113$, find the value of p , giving your answer correct to 2 decimal place. [2]
- Assume that $p = 0.345$.
- (i) Find the probability that in a random sample of 60 items, there are more than 20 items that are faulty. [2]
 - (ii) Using a suitable approximation, find the probability that in a sample of 100 randomly chosen items, the number of faulty items lies between 20 and 25 (inclusive). [4]

10. (a) Explain why it is advisable to plot a scatter diagram before interpreting a correlation coefficient calculated for a sample drawn from a bivariate distribution. [1]

Sketch two scatter diagrams indicating the following

- (i) two variables having a strong, negative linear correlation;
 (ii) two variables having a weak, positive linear correlation. [2]

- (b) A teacher wants to research on the correlation between the weight JJ students and the mean amount of money spent on canteen food per month by JJ students. Eight JJ students were interviewed and had their weights taken. The results are tabulated as follows:

Weight, x / kg	36	45	69	72	59	53	33	48
Mean amount spent on food per month, y / \$	54	60	84	91	79	k	49	68

- (i) Given that the equation of the least squares regression line of y on x is $y = 1.0205x + 16.0617$, show that the value of k is 67. [3]
- (ii) Give a sketch of the scatter diagram for the data as shown on your calculator. [2]
- (iii) Calculate the value of the linear product moment correlation coefficient, r , for these eight students. [1]
- (iv) State with a reason, the effect on r if the prices of food in the canteen have increased. [1]
- (v) The principal uses the regression line in part (i) to predict the weight of the student if the student spent a mean amount of \$80 on food per month, Comment briefly on the reliability of the estimate. [2]

- 11.** The Land Transport Authority (LTA) wishes to find out the average amount each Singaporean spends on public transport per month. A list of Singaporeans, living in a particular residential estate, is generated and every 50th Singaporeans on the list will be interviewed to obtain a sample of 400 responses.

- (i) Explain why the sample is biased. [1]
- (ii) What type of sampling method is being used? [1]
- (iii) If the LTA is interested to find out the responses of Singaporeans, by the different age groups between the ages of 13 and 50, suggest a better sampling method and justify your method. [2]

The LTA believes that each Singaporean spends an average of \$145 per month on public transport. A random sample of 400 Singaporeans was interviewed and their expenditure, in \$ x , is summarized as follows:

$$\sum (x - 200) = -22800 \quad \text{and} \quad \sum (x - \bar{x})^2 = 299780$$

- (iv) Calculate the unbiased estimates of the population mean μ and variance σ^2 , giving your answer correct to 2 decimal places. [2]
- (v) Test, at the 4% level of significance, whether the sample provides significant evidence that the LTA has overestimated the average amount Singaporeans spends on public transport per month. [4]
- (vi) Find the smallest level of significance at which the test would result in rejection of the claim of the LTA. [2]
- (vii) Explain what do you understand by the phrase “at the 4% level of significance” in the context of this question. [1]

12. A fruit orchard is well known for its oranges and apples. The mass of an orange and the mass of an apple are modeled as having independent normal distributions with the following means and standard deviations.

	Mean	Standard deviation
Mass of an orange	280 grams	22 grams
Mass of an apple	220 grams	18 grams

The price of oranges is \$2 per kilogram and the price of apples is \$3 per kilogram.

- (i) Find the probability that two oranges randomly chosen from the orchard have a total mass of more than 620 grams. [3]
- (ii) If random samples of 80 oranges and 100 apples are chosen, find the probability that the mean mass of oranges differs from the mean mass of apples by less than 56 grams. [4]
- (iii) The orchard wishes to advertise that at least 98% of its oranges have masses of at least m grams. Find the maximum value of m for the advertisement to be truthful, if m is to be an integer. [3]
- (iv) Find the probability that the selling price of a randomly chosen apple is less than \$0.60. [4]

-----**The End**-----